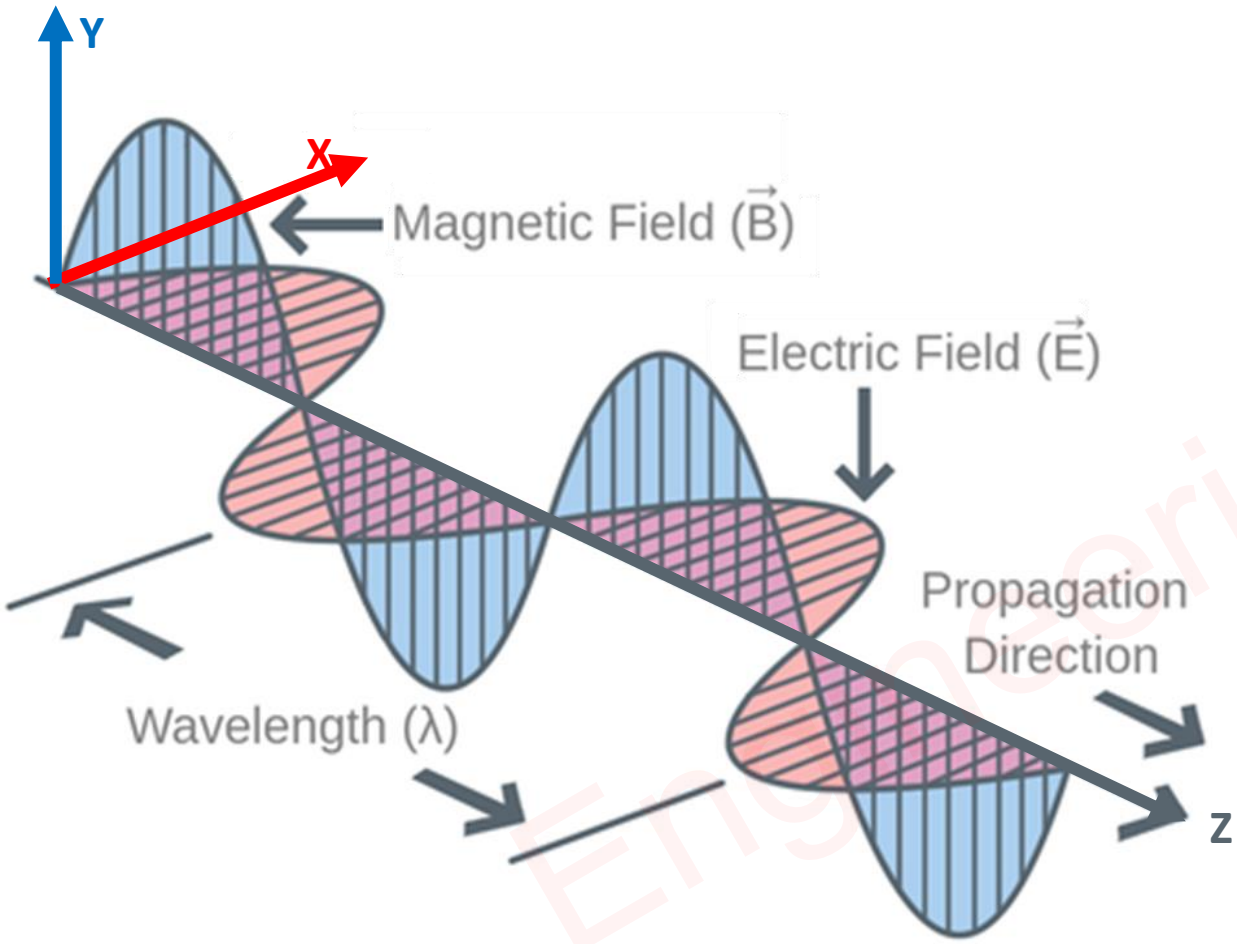
EM Waves in Free Space & Medium



- Direction of EM wave is $\vec{E} \times \vec{H}$.

For Free Space

- For EM wave, Electric Field is given by:

$$\Longrightarrow \vec{E} = E_0 \sin(\omega t - \beta z)\, \hat{i}_x$$

- Magnetic Field Intensity is given by:

$$\Longrightarrow \vec{H} = H_0 \sin(\omega t - \beta z)\, \hat{i}_y$$

For Medium Space

- For EM wave, Electric Field is given by:

$$\Longrightarrow \vec{E} = E_0 e^{-\alpha Z} \sin(\omega t - \beta z)\, \hat{i}_x$$

- Magnetic Field Intensity is given by:

$$\Longrightarrow \vec{H} = H_0 e^{-\alpha Z} \sin(\omega t - \beta z)\, \hat{i}_y$$

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Wave Equation of EM Waves in Free Space

~) As per Faraday's Law [Point Form]

$$\Rightarrow \nabla \times E = -\frac{\partial B}{\partial t}$$

~) Take Curl on both sides.

$$\Rightarrow \nabla \times \nabla \times E = -\frac{\partial (\nabla \times B)}{\partial t}$$

$$\Rightarrow \nabla(\nabla \cdot E) - \nabla^2 E = -\frac{\partial (\nabla \times B)}{\partial t} \quad \text{—} (1)$$

~) As per Gauss Law of Electric field [Point Form]

$$\Rightarrow \nabla \cdot D = \rho_v$$

$$\Rightarrow \nabla \cdot E = \rho_v / \varepsilon_0$$

- In Free space $\rho_v = 0$

$$\Rightarrow \nabla \cdot E = 0 \quad \text{—} (2)$$

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→ Put ② into ①.

$$\Rightarrow \nabla^2 E = \frac{\partial(\nabla \times B)}{\partial t} \quad \text{— (A)}$$

→ As per Ampere circuit Law for time varying field [Point Form].

$$\Rightarrow \nabla \times H = J_c + \frac{\partial D}{\partial t} \quad \text{— (3)}$$

→ For Free Space $J_c = 0$

$$\Rightarrow \nabla \times H = \frac{\partial D}{\partial t}$$

$$\Rightarrow \nabla \times B = \mu_0 \varepsilon_0 \frac{\partial E}{\partial t} \quad \text{— (4)}$$

→ Put eq.^{n} (4) into eq.^{n} (A).

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$$\Rightarrow \nabla^2 E = \mu_0 \varepsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$\rightsquigarrow$ For Free Space, $c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$

$$\Rightarrow \boxed{\nabla^2 E = \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2}} \quad \text{— (A)}$$

$$= \nabla^2 E = \frac{1}{c^2} \omega^2 E = \mu \varepsilon \omega^2 E$$

$$\Rightarrow \boxed{\nabla^2 E = \gamma^2 E} \quad \text{— (B)}$$

$\rightsquigarrow$ where, γ = propagation constant

$\rightsquigarrow$ eqn (A) and (B) are wave eqn.

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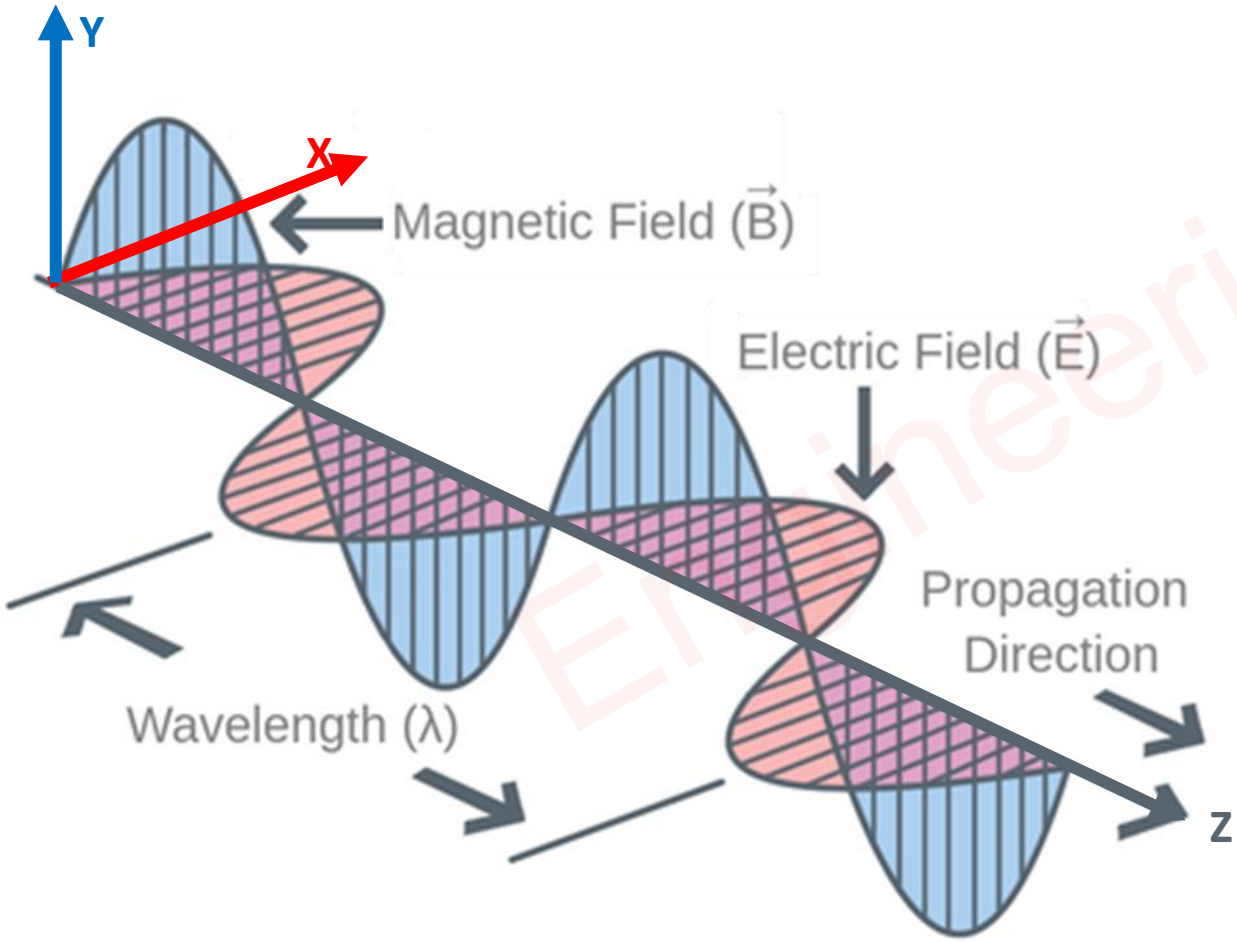
Propagation Constant, Attenuation Constant, and Phase Constant of Plane EM Waves



→ As per wave eq.^{n}

$$\Rightarrow \nabla^2 E = \gamma^2 E$$

→ γ propagation constant

$$\Rightarrow \gamma = \alpha + j\beta$$

$$= \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$$

→ Sol^n

$$\Rightarrow \alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}} - 1\right]^{1/2}$$

$$\Rightarrow \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}} + 1\right]^{1/2}$$

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For Lossless Medium

- For Lossless Medium, there is no loss of signal during propagation.
- Loss of signal happens due to the charge carrier absorbing the energy of EM waves.
- There is no free charge carrier in the lossless medium [$\sigma = 0$].

$$\Rightarrow \alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}} - 1\right]^{1/2} = 0$$

$$\Rightarrow \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}} + 1\right]^{1/2} = \omega\sqrt{\mu\varepsilon}$$

$$\rightarrow \gamma = \alpha + j\beta$$

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For Lossy Medium

- For Lossy Medium, there is a loss of signal during propagation.
- Loss of signal happens due to the charge carrier absorbing the energy of EM waves.

$$\Rightarrow \frac{\sigma}{\omega\varepsilon} \gg 1$$

$$\Rightarrow \alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}} - 1\right]^{1/2} = \omega\sqrt{\frac{\mu\varepsilon}{2}}\sqrt{\frac{\sigma}{\omega\varepsilon}} = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\Rightarrow \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}} + 1\right]^{1/2} = \omega\sqrt{\frac{\mu\varepsilon}{2}}\sqrt{\frac{\sigma}{\omega\varepsilon}} = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\rightarrow \gamma = \alpha + j\beta$$

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Skin Effect

Outlines

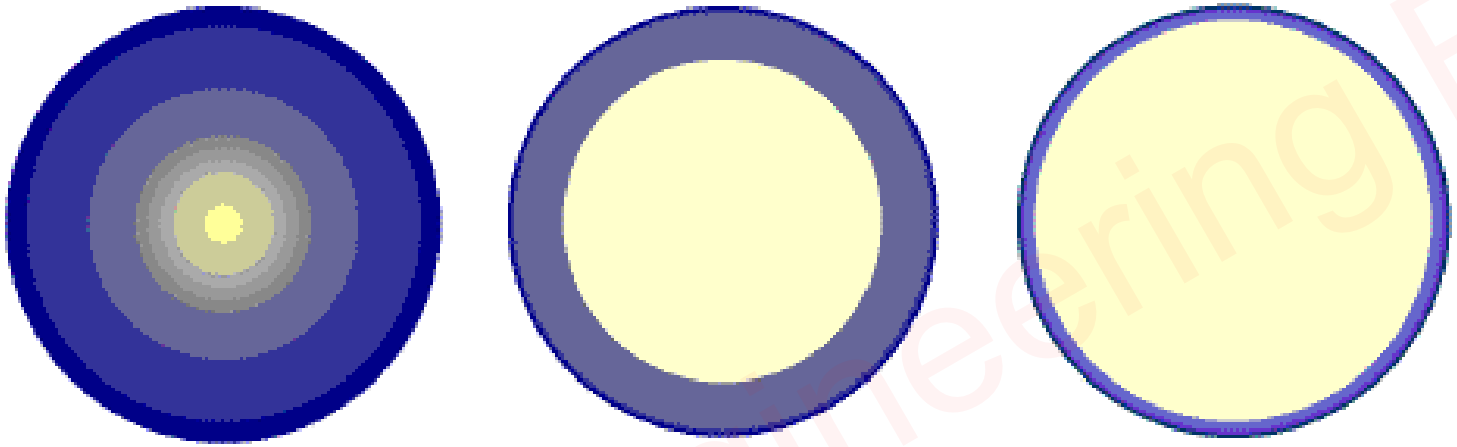
- Basics of Skin Effect
- Derivation of Skin Depth
- Parameters of Skin Depth

Engineering Funda

Skin Effect

Basics of Skin Effect

- Skin Effect explains that The Conductive Energy of EM wave stays on the skin (Surface) of the conductive Material.



1KHz 100KHz 1000KHz

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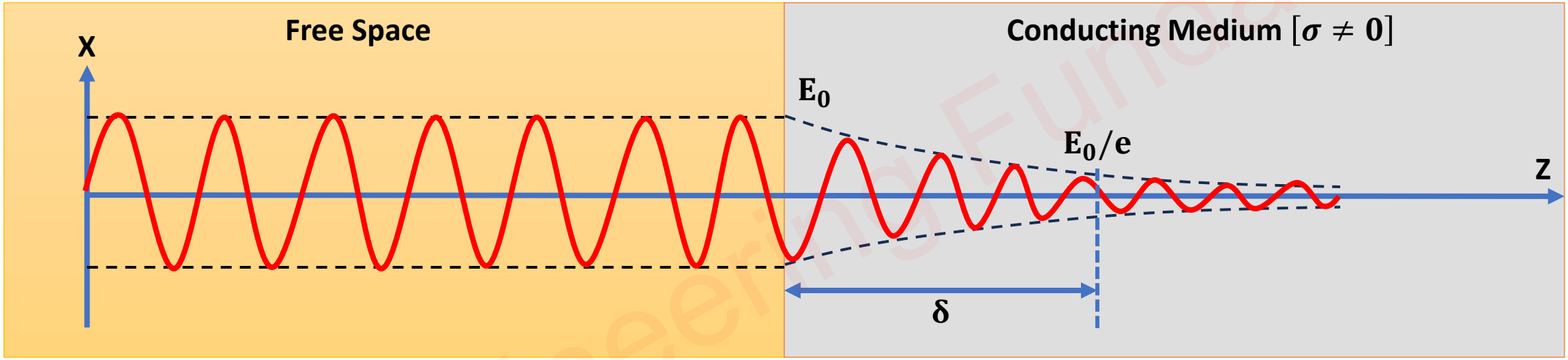
Derivation of Skin Depth

- For EM wave, Electric Field is given by:

$$\Longrightarrow \vec{E} = E_0 e^{-\alpha z} \sin(\omega t - \beta z)\, \hat{i_x}$$



- When amplitude of signal decreases to $(1/e)^{th}$ of the it's initial amplitude at certain depth δ, that depth is Skin Depth δ.

$$\Longrightarrow E_0 e^{-\alpha\delta} = E_0/e$$

$$\Longrightarrow e^{+\alpha\delta} = e$$

$$\Longrightarrow \delta = 1/\alpha$$

- Attenuation Constant $\alpha = \sqrt{\frac{\mu\sigma\omega}{2}}$

$$\Longrightarrow \delta = \sqrt{\frac{2}{\mu\sigma\omega}}$$

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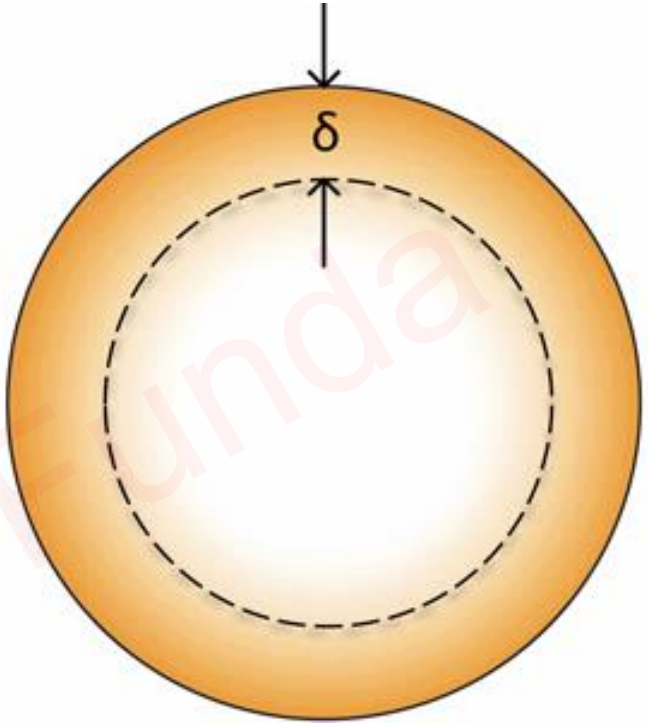
Parameters of Skin Depth

- Skin Depth of Conductive Material is given by:

$$\Rightarrow \delta = \sqrt{\frac{2}{\mu\sigma\omega}}$$

- Conductivity σ = 1 / (Resistivity ρ)

$$\Rightarrow \delta = \sqrt{\frac{2\rho}{\mu\omega}}$$



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Intrinsic Wave Impedance

Outlines

- Basics of Intrinsic Wave Impedance
- Intrinsic Wave Impedance of Lossless Medium
- Intrinsic Wave Impedance of Lossy Medium

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Basics of Intrinsic Wave Impedance

- Intrinsic Wave Impedance is the impedance of the medium, in which EM wave is travelling.
- So based on the medium, we can define Intrinsic Wave Impedance as per:

$$\Rightarrow \eta = \frac{E}{H}$$

- Intrinsic Wave Impedance can also be calculated as per:

$$\Rightarrow \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

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Intrinsic Wave Impedance of Lossless Medium

- Lossless Medium has no loss of EM wave.
- For Lossless Medium, Free Charge carriers are zero $[\sigma = 0]$.

$$\Rightarrow \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$\Rightarrow \eta = \sqrt{\frac{j\omega\mu}{j\omega\varepsilon}}$$

$$\Rightarrow \eta = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0\mu_r}{\varepsilon_0\varepsilon_r}}$$

[∴ For free space, $\mu_r = 1$ and $\varepsilon_r = 1$]

$$\Rightarrow \eta = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}}}$$

$$\Rightarrow \eta = 377\Omega = 120\pi\Omega$$

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Intrinsic Wave Impedance of Lossy Medium

- For Lossy Medium, It has a loss of EM waves due to charge carriers $[\sigma \gg 1]$.

$$\Rightarrow \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$\Rightarrow \eta = \sqrt{\frac{j\omega\mu}{\sigma}}$$

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Loss Tangent of Medium

☆ Basics of Loss Tangent

- Loss Tangent is angle of loss for given medium.
- Loss tangent depends on loss of EM wave in the given medium.

☆ Derivation of Loss Tangent

- As per Ampere Circuit Law [point Form].

$$\Rightarrow \nabla \times H = J_c + J_d$$

$$\Rightarrow \nabla \times H = J_c + \frac{\partial D}{\partial t} \qquad [\, D = \varepsilon E \,]$$

$$\Rightarrow \nabla \times H = \underbrace{J_c}_{\text{loss component}} + \underbrace{\varepsilon \frac{\partial E}{\partial t}}_{\text{lossless component}}$$

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→ Loss Tangent

$$\Rightarrow \tan\theta_l = \frac{J_c}{J_d} = \frac{\sigma E}{\varepsilon\omega E} = \frac{\sigma}{\varepsilon\omega} \quad \text{— (A)}$$

★ Relation of Intrinsic Impedance angle θ_n with loss Tangent.

$$\Rightarrow 2\theta_n = \theta_l$$

→ From eq.^{n} (A).

$$\Rightarrow \tan(2\theta_n) = \frac{\sigma}{\varepsilon\omega}$$

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Poynting Theorem

- Poynting Theorem is used to identify the Power Density {Rate of Energy per unit Area} of EM waves.

~ Power Density S.

$$\Rightarrow \vec{S} = \vec{E} \times \vec{H} \quad \text{—①} \quad \left[\frac{V}{m} \times \frac{A}{m} = \frac{W}{m^2} = \frac{E}{t\,m^2} \right]$$

~ EM waves.

$$\left.\begin{aligned} \Rightarrow \vec{E} &= E_0 e^{-\alpha z} \sin(\omega t - \beta z + \theta_1)\, \hat{i}_x \\ \Rightarrow \vec{H} &= H_0 e^{-\alpha z} \sin(\omega t - \beta z + \theta_2)\, \hat{i}_y \end{aligned}\right\} \text{—②}$$

~ Put ② into ①.

$$\Rightarrow \vec{S} = E_0 e^{-\alpha z} \sin(\omega t - \beta z + \theta_1)\, \hat{i}_x \times H_0 e^{-\alpha z} \sin(\omega t - \beta z + \theta_2)\, \hat{i}_y$$

$$\Rightarrow \vec{S} = E_0 H_0 e^{-2\alpha z} \sin(\omega t - \beta z + \theta_1) \sin(\omega t - \beta z + \theta_2)\, \hat{i}_z$$

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$$\rightsquigarrow S_{avg} = \frac{1}{T}\int_0^T S\,dt$$

$$= \frac{1}{T}\int_0^T E_0 H_0 e^{-2\alpha z} \sin(\omega t - \beta z + \theta_1)\sin(\omega t - \beta z + \theta_2)\,dt$$

$$= \frac{E_0 H_0}{T} e^{-2\alpha z} \int_0^T \frac{1}{2}\left[\cos(\theta_1 - \theta_2) - \cos(2\omega t - 2\beta z + \theta_1 + \theta_2)\right] dt$$

$$= \frac{E_0 H_0}{\cancel{T}} e^{-2\alpha z} \left[\frac{1}{2}\cos(\theta_1 - \theta_2)\right]\cancel{T}$$

$$= \frac{E_0 H_0}{2} e^{-2\alpha z} \cos(\theta_1 - \theta_2)$$

$$\rightarrow \eta = \frac{E_0 \angle\theta_1}{H_0 \angle\theta_2} = \frac{E_0}{H_0}\angle\theta_1 - \theta_2 = \frac{E_0}{H_0}\angle\theta_\eta, \quad \text{where } \theta_\eta = \theta_1 - \theta_2$$

= Intrinsic Impedance angle.

$$\boxed{\Rightarrow S_{avg} = \frac{E_0 H_0}{2} e^{-2\alpha z} \cos\theta_\eta}$$

← lossy Medium.

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→ For lossless medium, $\alpha = 0$, $Q_n = 0$

$$\boxed{\Rightarrow S_{avg} = \frac{E_0 H_0}{2} = \frac{E_0^2}{2\eta}}$$ ← lossless medium

→ Average power P_{avg}.

$$\Rightarrow P_{avg} = \int S_{avg}\, ds$$

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Example of EM Waves

Question 1: The time domain expression for the magnetic field of a plane wave traveling in a Non-magnetic medium is given by:

↳ $\mu_r = 1$

direction x

$$H(x,t) = \underline{0.5}\cos(\underbrace{6\pi \times 10^8}_{\omega = 6\pi \times 10^8 \text{ rad/sec}} t - \underbrace{10}_{\beta = 10 \text{ rad/m}} x)\,\widehat{a_z}\ (A/m)$$

A. Find the direction of wave propagation
B. The dielectric constant and intrinsic impedance
C. The time domain expression of the Electric Field
D. Average Power Density

→ Direction of EM $= \vec{E} \times \vec{H}$

$\hat{a}_x = \vec{E} \times \hat{a}_z$

A) - x - direction.

B) - $\beta = \omega\sqrt{\mu\varepsilon} = \omega\sqrt{\mu_0\mu_r\varepsilon_0\varepsilon_r}$

$\Rightarrow 10 = 6\pi \times 10^8 \sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12} \times \varepsilon_r}$

$\Rightarrow \sqrt{\varepsilon_r} = \dfrac{10}{6\pi \times 10^8} \times 3 \times 10^8 = 1.59$

$\Rightarrow \varepsilon_r = 2.53$

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- $\eta = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0 \mu_r}{\varepsilon_0 \varepsilon_r}}$

$= \sqrt{\frac{4\pi \times 10^{-7} \times 1}{8.854 \times 10^{-12} \times 2.53}}$

$= 236\ \Omega$

- $\eta = \frac{E}{H} \Rightarrow E = \eta H$

$= 236 \times 0.5$

$= 118\ V/m$

c) $E = 118 \cos(6\pi \times 10^{8} t - 10x)\ \hat{a}_y\ V/m$

d) $S = \frac{E^2}{2\eta} = \frac{118^2}{2 \times 236} = 29.5\ W/m^2$

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Examples of EM Waves

Question 2: A monochromatic plane EM waves travel in a vacuum in the positive x direction. The electric and magnetic field expression can be expressed as ……..

~ EM wave $\longrightarrow$ x direction.

~ Direction of EM wave $= \vec{E} \times \vec{H}$

$$\Rightarrow \hat{a}_x = \hat{a}_y \times \hat{a}_z$$
$$= \hat{a}_z \times (-\hat{a}_y)$$
$$= (-\hat{a}_z) \times \hat{a}_y$$

~ For E field

$$\vec{E} = E_0 \sin(\omega t - kx)\, \hat{a}_y$$

~ For H field

$$\vec{H} = H_0 \sin(\omega t - kx)\, \hat{a}_z$$

(1)

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→ For E field

$$\vec{E} = E_0 \sin(\omega t - kx)\,\hat{a}_z$$

→ For H field

$$\vec{H} = -H_0 \sin(\omega t - kx)\,\hat{a}_y$$

(2)

→ For E field

$$\vec{E} = -E_0 \sin(\omega t - kx)\,\hat{a}_z$$

→ For H field

$$\vec{H} = H_0 \sin(\omega t - kx)\,\hat{a}_y$$

(3)

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Question 3: In an Electromagnetic wave in free space, the root mean square value of the electric field is 6 V/m. The peak value of the magnetic field is ....2.82×10^{-8} T

$\rightsquigarrow$ $E_{rms} = 6$ V/m

$\rightsquigarrow$ $\eta = 120\pi = \dfrac{E_{rms}}{H_{rms}}$

$\Rightarrow$ $H_{rms} = \dfrac{E_{rms}}{120\pi} = \dfrac{6}{120\pi} = 0.0159$ A/m

$\rightsquigarrow$ $B_{rms} = \mu H_{rms}$

$= \mu_0 \mu_r H_{rms}$

$= 4\pi \times 10^{-7} \times 1 \times 0.0159$

$= 2 \times 10^{-8}$ T

$\rightsquigarrow$ $B_m = \sqrt{2}\, B_{rms}$

$= 2.82 \times 10^{-8}$ T

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Question 4: The velocity of electromagnetic radiation in a medium of permittivity ε_0 and Permeability μ_0 is given by ....3×10^8 m/sec

$\varepsilon = \varepsilon_0 \varepsilon_r$

$\mu = \mu_0 \mu_r$

$$\Rightarrow c = \frac{1}{\sqrt{\mu \varepsilon}} = \frac{1}{\sqrt{\mu_0 \mu_0 \varepsilon_0 \varepsilon_r}} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \text{ m/sec.}$$

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